Nandi Gabriella Shandu

nandishandu51@gmail.com 0743142722 Durban, KwaZulu Natal 2003-10-31

linkedin.com/in/nandi-gabriella-shandu-079231384 github.com/Nandi288

Summary

Final-year ICT Application Development student at Durban University of Technology, with a strong record of collaboration in Agile sprint teams and hackathons. Builds secure, scalable web applications with C#, ASP.NET Core MVC, Entity Framework Core and SQL Server, and deploys them on Microsoft Azure. Brings clean, maintainable code and the ability to deliver under tight deadlines to a remote software engineering team.

Professional Experience

Geeked-On x Tech Horizons IT Incubation Program 06/2026 – 07/2026
Durban, KZN

- Developed full stack web applications using the ASP.NET MVC framework and C#, ensuring strict adherence to architectural patterns and clean separation of concerns.
- Worked within a sprint style team environment of six developers to build enterprise software from scratch for a Demo Day milestone, using Git for version control and GitHub for repository management and team code merging.
- Used AI tools and prompt engineering methodologies to accelerate frontend UI construction and optimize backend business logic pipelines, reducing development time across a 6 member sprint team building an enterprise grade web application from scratch for a public Demo Day milestone.

Participant, *KZN Imbokodo Hackathon*

- Conducted user centric research on menstrual equity and female entrepreneurship to define problem statements, buyer and seller personas, and B2B business models for local micro enterprises. This research became the foundation for the system design work now detailed under the HerCycle project.
- Participated in an intensive 24 plus hour coding sprint, collaborating closely with teammates and industry mentors to deliver a full product pitch and system demonstration.

Geekulcha Hackathon 2026 Participant (Backend Developer)

- Selected to compete in an intensive hackathon sprint following qualification through the Geeked-On x Tech Horizons Incubation Programme.
- Collaborated with a multidisciplinary team to architect scalable ASP.NET Core backend services and design relational database schemas under tight sprint deadlines.
- Managed version control and branch integration workflows using Git and GitHub in a fast paced team environment.

FNB App of the Year Hackathon 2026

• Participated in the FNB App of the Year Hackathon 2026 under the Youth Employment and Work Readiness challenge, presenting Indlela to judges with a live demo and pitch.
• Worked with a team to design the product, build the app, and prepare the submission.

Education

Diploma in ICT: Application Development, *Durban University of Technology* 01/2024 – 11/2026
Durban ,
Kwazulu Natal

National Senior Certificate, *J.G Zuma High School* 01/2017 – 11/2021
Durban ,
Kwazulu Natal

Skills

Languages
- C#
- JavaScript
- SQL
- HTML5
- CSS3

Tools and Methodologies
- Git
- GitHub
- MVC Architecture
- Agile/Scrum Principles
- Technical Documentation

Backend and Frameworks
- ASP.NET Core MVC
- ASP.NET MVC,
- .NET 8
- Entity Framework Core

Databases
- SQL Server
- Azure Cosmos DB
- Azure Blob Storage

Frontend Development
- React.js

Certificates

Full Stack Development
FNB Academy 2025

Introduction to Cybersecurity
Cisco Networking Academy

Projects

Student Management System
Technologies used: C#, ASP.NET Core MVC, .NET 8, Microsoft Azure, Azure Cosmos DB, Azure Blob Storage, OAuth 2.0

- Built and deployed a full-stack student management portal on Microsoft Azure within a collaborative 10 person development team.
- Engineered an OAuth 2.0 authentication pipeline (GitHub and Google) paired with Role-Based Access Control (RBAC), securing administrative endpoints by enforcing a strict allow list policy.
- Implemented full CRUD operations handling multi-attribute student records, reducing administrative processing bottlenecks.
- Isolated sensitive asset storage by querying Azure Cosmos DB for structured textual data while routing profile images to Azure Blob Storage protected by time limited Shared Access Signature (SAS) tokens.

Security Management System (Pirana Guarding)
Technologies used: C#, ASP.NET MVC, SQL Server, HTML5, CSS3, JavaScript

- Developed an operational guard tracking web portal engineered to manage daily shift rotations for up to 12 active guards per site.
- Engineered an anti-fraud check-in engine using dynamically generated, daily expiring QR codes, eliminating 100% of static credential and code-reuse attendance tampering.
- Built automated business rule validation that evaluates guard shift statuses in real time, immediately blocking unauthorized login attempts for off-duty personnel.
- Designed a centralized Director Dashboard displaying real-time attendance logs, active timestamps, and incident reporting metrics.

EcoRewards SA — Smart Recycling Rewards Web Application
Technologies used: C#, ASP.NET MVC, .NET Framework 4.8, Entity Framework 6, SQL Server, Bootstrap 5, Chart.js

- Architected a Code-First relational database schema in SQL Server via Entity Framework 6 to manage user profiles, waste categories, point balances, and reward items.
- Built role-based access control via ASP.NET Identity across 3 discrete user roles (Resident, Municipal Staff, Administrator) with tailored permissions and views.
- Programmed automated reporting engines using iTextSharp and EPPlus, allowing administrators to generate and export custom monthly/annual PDF and Excel data summaries.
- Integrated interactive Chart.js dashboards to track monthly recycling trends, collection event metrics, and reward redemptions.
- Pitched and demonstrated the live software prototype to industry judges and stakeholders at Demo Day.

HerCycle-Digital Marketplace for Women Entrepreneurs in Menstrual Wellness
Technologies used: ASP.NET Core MVC, SQL Server, Entity Framework Core, ASP.NET Core Identity, Bootstrap 5, Chart.js

- Developed and presented a functional e-commerce marketplace during a 24-hour hackathon sprint, transitioning micro-entrepreneurs from informal WhatsApp sales to a centralized web platform.
- Engineered an end-to-end seller management suite featuring real-time inventory tracking, order status workflows, and interactive sales analytics.
- Built customer discovery workflows including multi-category filtering, shopping cart state management, wishlists, and checkout pipelines.
- Implemented a dual-tier subscription architecture using ASP.NET Core Identity, enforcing listing limits between Free (5 items) and Premium (50 items at R99/month) accounts.

VisionScribe — AI-Powered Image Analysis Application
Technologies used: C#, ASP.NET Core MVC, .NET 8, Azure Computer Vision API, Azure Blob Storage, IMemoryCache, REST APIs

- Architected a full-stack image intelligence application utilizing ASP.NET Core MVC (.NET 8) to parse, tag, and synthesize written descriptions from uploaded photos within seconds.
- Integrated the Azure Computer Vision REST API with Azure Blob Storage pipelines, transforming raw vision payloads (captions, dominant colors, detected objects) into structured user insights.
- Engineered an in-memory caching layer using IMemoryCache for repeated uploads, eliminating 100% of duplicate Azure Cognitive API calls and significantly cutting network latency.
- Applied SOLID principles and Dependency Injection by decoupling IBlobStorageService and IVisionAnalysisService interfaces from controllers to ensure testability and modularity.

Indlela: Free Learning-to-Employment Platform
Technologies used: C#, ASP.NET Core 8 MVC, Entity Framework Core, SQL Server, Azure SQL, ASP.NET Core Identity, Microsoft Azure

• Built and deployed a live web app that helps young South Africans learn free courses, prove their skills, and apply for matching jobs.

• Designed the learner journey: email verification with secure sign-in, a Smart Start roadmap recommender, a course catalogue, a progress dashboard, and skill-based job matching.
• Built a one-column ATS-friendly CV generator, QR-verified certificates with short codes, and a public verification page.
• Added a background service that refreshes South African tech news from RSS feeds every six hours.